10th Grade Midterm Analysis of Decisions: C7 Solutions

Contents

1 Introduction 1

2 Contextualized 2x2 Problem - Clothing store 6

3 2x2 - Coffee shop 7

4 2x2 - Manufacturing company 9

5 2x2 - Retail company 11

6 2x2 - Retail company 12

7 2x3 - Delivery company 14

8 2x3 - Manufacturing company 16

9 3x2 - Technology company 18

10 3x2 - Renewable energy company 21

11 3x2 - Restaurant chain 22

12 3x2 - Agricultural cooperative 23

Evaluated Criterion

C7: Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

1 Introduction

Problems:

1.1. Expected Value and Sensitivity Analysis

1.2. General 2x2 Problem

1.3. General 2x3 Problem

1.4. General 3x2 Problem

1.1 Expected Value and Sensitivity Analysis

When probabilities are subjective or estimated, a decision-maker can examine how changes in the probability assumptions affect the preferred alternative. The Expected Value criterion allows each alternative to be represented as a function of the probability assigned to the possible states of nature. For a payoff table with alternatives in rows and states of nature in columns, let $P_{ay}(A_i, S_j)$ represent the payoff obtained when alternative A_i is selected and state of nature S_j occurs.

Let p represent the subjective probability that state of nature S_1 occurs. Since there are two possible states,

$$P(S_1) = p \qquad \text{and} \qquad P(S_2) = 1 - p \qquad \text{with} \quad 0 \leq p \leq 1$$

For each alternative, calculate its Expected Value as a function of p:

$$EV(A_i) = pP_{ay}(A_i, S_1) + (1 - p)P_{ay}(A_i, S_2).$$

This expression shows how the expected payoff changes when the subjective probability of S_1 changes. The preferred alternative depends on the value of p. To determine which alternative is preferred, compare their expected-value functions. For two alternatives A_1 and A_2, the threshold probability is found by solving

$$EV(A_1) = EV(A_2).$$

At this threshold, the decision-maker is indifferent between the two alternatives. The preferred alternative on each side of the threshold is determined by comparing the corresponding expected values. Therefore, sensitivity analysis identifies how changes in the subjective probability affect the preferred decision.

1.2 General 2x2 Problem

Consider the following payoff table, where the probability of state S_1 is p and the probability of state S_2 is $1 - p$:

Alternative	S_1 (p)	S_2 ($1 - p$)
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

Determine the preferred alternative as a function of p, including the threshold probability at which the decision-maker is indifferent.

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1, A_2 States of nature: S_1, S_2

Probabilities: $P(S_1) = p$ $P(S_2) = 1 - p$

Step 2: Calculate the expected value of A_1

For alternative A_1:

$$EV(A_1) = pP_{ay}(A_1, S_1) + (1-p)P_{ay}(A_1, S_2).$$

This expression shows how the expected payoff of A_1 changes as p changes.

Step 3: Calculate the expected value of A_2

For alternative A_2:

$$EV(A_2) = pP_{ay}(A_2, S_1) + (1-p)P_{ay}(A_2, S_2).$$

This expression shows how the expected payoff of A_2 changes as p changes.

Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2).$$

Solve this equation for p. The resulting value is the probability at which the decision-maker is indifferent between the two alternatives.

Step 5: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$$\begin{array}{lcl} EV(A_1) > EV(A_2) & \Rightarrow & \text{choose } A_1, \\ EV(A_2) > EV(A_1) & \Rightarrow & \text{choose } A_2, \\ EV(A_1) = EV(A_2) & \Rightarrow & \text{indifferent.} \end{array} \quad \longrightarrow \quad \textbf{Decision rule} \quad \begin{cases} A_1, & \text{if } EV(A_1) > EV(A_2), \\ A_2, & \text{if } EV(A_2) > EV(A_1), \\ \text{indifferent}, & \text{if } EV(A_1) = EV(A_2). \end{cases}$$

Thus, the preferred alternative can be expressed as a function of the subjective probability p.

Extended payoff table

Alternative	S_1 (p)	S_2 $(1-p)$	Expected Value
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$pP_{ay}(A_1, S_1) + (1-p)P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$pP_{ay}(A_2, S_1) + (1-p)P_{ay}(A_2, S_2)$

1.3 General 2x3 Problem

Consider the following payoff table, where the probabilities of the states of nature are subjective and may vary:

Alternative	S_1 (p)	S_2 (q)	S_3 $(1-p-q)$
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$

$$P(S_1) = p, \qquad P(S_2) = q, \qquad P(S_3) = 1 - p - q$$

$$p \geq 0, \qquad q \geq 0, \qquad p + q \leq 1$$

Determine the preferred alternative as a function of the subjective probabilities p and q, including any threshold values or boundaries at which the preferred decision changes.

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1, A_2 States of nature: S_1, S_2, S_3

Probabilities: $P(S_1) = p, \quad P(S_2) = q, \quad P(S_3) = 1 - p - q.$

Step 2: Calculate the expected value of A_1

$$EV(A_1) = pP_{ay}(A_1, S_1) + qP_{ay}(A_1, S_2) + (1 - p - q)P_{ay}(A_1, S_3).$$

This expression shows how the expected payoff of A_1 changes as the subjective probabilities change.

Step 3: Calculate the expected value of A_2

$$EV(A_2) = pP_{ay}(A_2, S_1) + qP_{ay}(A_2, S_2) + (1 - p - q)P_{ay}(A_2, S_3).$$

Step 4: Compare the expected-value functions and perform the sensitivity analysis

Compare

$$EV(A_1), \quad EV(A_2).$$

The preferred alternative depends on the values of p and q. Examine the preferred alternative for different feasible combinations of p and q, subject to

$$p \geq 0, \qquad q \geq 0, \qquad p + q \leq 1.$$

The boundary occurs when the expected values of the two alternatives are equal, indicating indifference:

$$EV(A_1) = EV(A_2) \quad \text{at a boundary.}$$

The preferred alternative is the one with the highest expected value:

$$A^* = \arg\max_i EV(A_i)$$

Extended payoff table

Alternative	S_1 p	S_2 q	S_3 $1 - p - q$	Expected Value
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$	$pP_{ay}(A_1, S_1) + qP_{ay}(A_1, S_2) + (1 - p - q)P_{ay}(A_1, S_3)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$	$pP_{ay}(A_2, S_1) + qP_{ay}(A_2, S_2) + (1 - p - q)P_{ay}(A_2, S_3)$

1.4 General 3x2 Problem

Consider the following payoff table, where the probabilities of the states of nature are subjective and may vary:

Alternative	S_1	S_2
	(p)	$(1-p)$
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$

$$P(S_1) = p, \qquad P(S_2) = 1 - p$$

$$0 \le p \le 1$$

Determine the preferred alternative as a function of the subjective probability p, including any threshold values at which the preferred decision changes.

Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1, A_2, A_3 States of nature: S_1, S_2

Probabilities: $P(S_1) = p, \qquad P(S_2) = 1 - p.$

Step 2: Calculate the expected value of A_1

$$EV(A_1) = pP_{ay}(A_1, S_1) + (1-p)P_{ay}(A_1, S_2).$$

This expression shows how the expected payoff of A_1 changes as the subjective probability changes.

Step 3: Calculate the expected value of A_2

$$EV(A_2) = pP_{ay}(A_2, S_1) + (1-p)P_{ay}(A_2, S_2).$$

Step 4: Calculate the expected value of A_3

$$EV(A_3) = pP_{ay}(A_3, S_1) + (1-p)P_{ay}(A_3, S_2).$$

Step 5: Compare the expected-value functions

Compare

$$EV(A_1), \quad EV(A_2), \quad EV(A_3).$$

Step 6: Perform the sensitivity analysis

The preferred alternative depends on the value of p, subject to

$$0 \le p \le 1.$$

For any two alternatives, a threshold occurs when their expected values are equal, indicating indifference:

$$EV(A_i) = EV(A_k) \quad \text{at a threshold.}$$

The preferred alternative is the one with the highest expected value:

$$A^* = \arg\max_i EV(A_i)$$

Extended payoff table

Alternative	S_1 p	S_2 $1-p$	Expected Value
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$pP_{ay}(A_1, S_1) + (1-p)P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$pP_{ay}(A_2, S_1) + (1-p)P_{ay}(A_2, S_2)$
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$	$pP_{ay}(A_3, S_1) + (1-p)P_{ay}(A_3, S_2)$

2 Contextualized 2x2 Problem - Clothing store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager is uncertain about the level of sales during the holiday season. Let p represent the subjective probability of strong sales. Therefore, the probability of weak sales is $1-p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine how the preferred decision changes as the subjective probability p changes. Identify the threshold probability at which the manager is indifferent between the two alternatives.

Alternative	Strong sales S_1 $P(S_1) = p$	Weak sales S_2 $P(S_2) = 1-p$
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	45	45

2.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Launch new campaign, A_2 = Keep current strategy

States of nature: S_1 = Strong sales, S_2 = Weak sales

Probabilities: $P(S_1) = p, P(S_2) = 1-p$ with $0 \leq p \leq 1$.

Step 2: Calculate the expected value of A_1

For the Launch new campaign alternative:

$$EV(A_1) = p(55) + (1-p)(5)$$

$$= 55p + 5 - 5p = 50p + 5.$$

Thus, the expected payoff from launching the new campaign is $50p + 5$ thousand dollars.

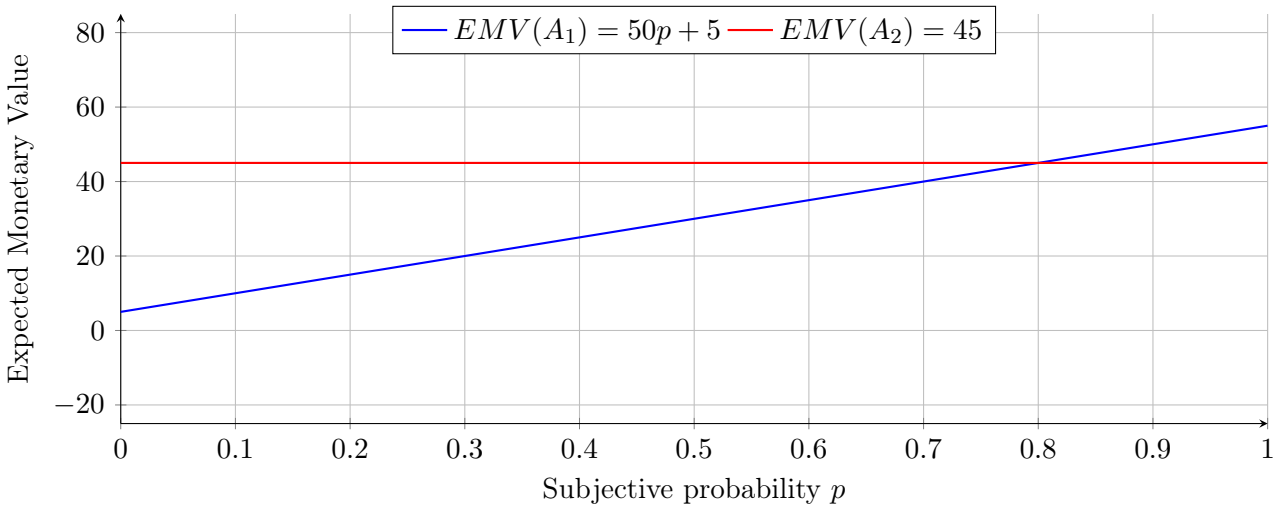
Step 3: Calculate the expected value of A_2

For the Keep current strategy alternative:

$$EV(A_2) = p(45) + (1-p)(45)$$

$$= 45p + 45 - 45p = 45.$$

Thus, the expected payoff from keeping the current strategy is 45 thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2) \quad \longrightarrow \quad 50p + 5 = 45$$
$$\longrightarrow \quad 50p = 40 \quad \longrightarrow \quad p = 0.8.$$

Thus, the manager is indifferent between the two alternatives when the subjective probability of strong sales is 80%.

Step 5: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$$\begin{array}{lcll} p < 0.800 & \Rightarrow & EV(A_1) < EV(A_2) & \Rightarrow \text{ choose } A_2, \\ p > 0.800 & \Rightarrow & EV(A_1) > EV(A_2) & \Rightarrow \text{ choose } A_1, \\ p = 0.800 & \Rightarrow & EV(A_1) = EV(A_2) & \Rightarrow \text{ indifferent.} \end{array} \quad \longrightarrow \quad \begin{cases} A_2, & 0 \le p < 0.800, \\ \text{indifferent}, & p = 0.800, \\ A_1, & 0.800 < p \le 1. \end{cases}$$

The sensitivity analysis shows that the clothing store should keep its current marketing strategy when the subjective probability of strong sales is below 80%. Only when that probability exceeds 80% does launching the new advertising campaign have the higher expected value.

Extended payoff table

Alternative	Strong sales S_1 $P(S_1) = p$	Weak sales S_2 $P(S_2) = 1 - p$	Expected Value
A_1: Launch new campaign	55	5	$p(55) + (1-p)(5) = 50p + 5$
A_2: Keep current strategy	45	45	$p(45) + (1-p)(45) = 45$

3 2x2 - Coffee shop

A small coffee shop is considering whether to introduce a new dessert line or keep its current menu. The manager is uncertain about the level of customer traffic during the promotional weekend. Let p represent the subjective probability

of high customer traffic. Therefore, the probability of low customer traffic is $1-p$. All payoffs are expressed in thousands of dollars. Using the Expected Value criterion, determine how the preferred decision changes as the subjective probability p changes. Identify the threshold probability at which the manager is indifferent between the two alternatives.

Alternative	High traffic S_1 $P(S_1) = p$	Low traffic S_2 $P(S_2) = 1-p$
Launch dessert line (A_1)	40	10
Keep current menu (A_2)	30	20

3.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Launch dessert line, A_2 = Keep current menu

States of nature: S_1 = High traffic, S_2 = Low traffic

Probabilities: $P(S_1) = p, P(S_2) = 1-p$ with $0 \le p \le 1$.

Step 2: Calculate the expected value of A_1

For the Launch dessert line alternative:

$$EV(A_1) = p(40) + (1-p)(10)$$

$$= 40p + 10 - 10p = 30p + 10.$$

Thus, the expected payoff from launching the dessert line is $30p + 10$ thousand dollars.

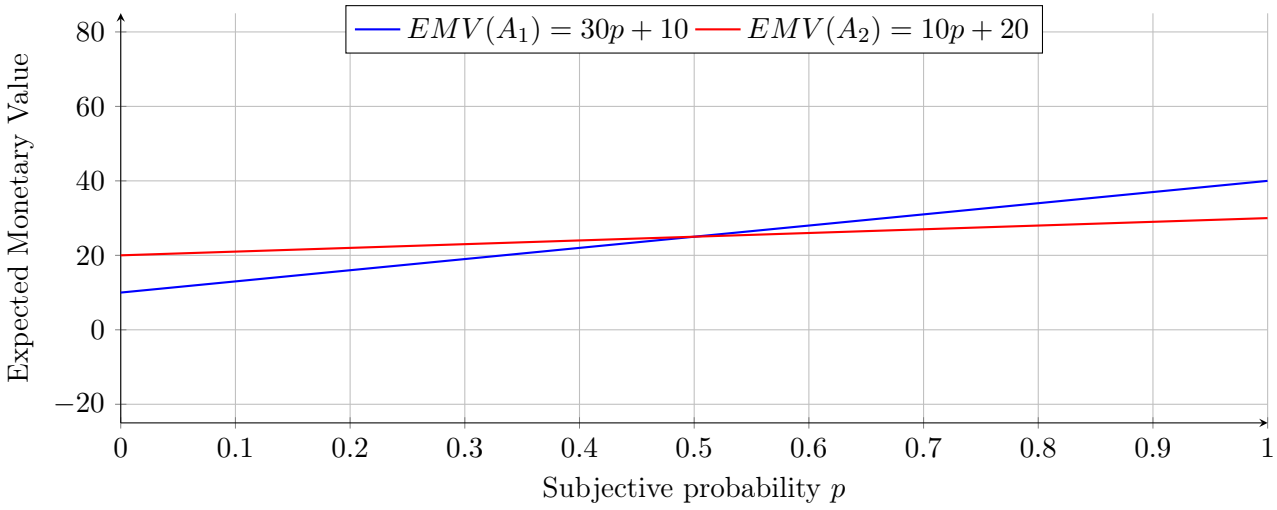
Step 3: Calculate the expected value of A_2

For the Keep current menu alternative:

$$EV(A_2) = p(30) + (1-p)(20)$$

$$= 30p + 20 - 20p = 10p + 20.$$

Thus, the expected payoff from keeping the current menu is $10p + 20$ thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2) \quad \longrightarrow \quad 30p + 10 = 10p + 20$$

$$\longrightarrow \quad 30p - 10p = 20 - 10 \quad \longrightarrow \quad 20p = 10 \quad \longrightarrow \quad p = \frac{10}{20} = \frac{1}{2} = 0.5$$

Thus, the manager is indifferent between the two alternatives when the subjective probability of high customer traffic is 50%.

Step 5: Perform the sensitivity analysis

Compare the expected values on each side of the threshold.

$$\begin{array}{lcllcl} p < 0.5 & \Rightarrow & EV(A_1) < EV(A_2) & \Rightarrow & \text{choose } A_2, & \\ p > 0.5 & \Rightarrow & EV(A_1) > EV(A_2) & \Rightarrow & \text{choose } A_1, & \longrightarrow \\ p = 0.5 & \Rightarrow & EV(A_1) = EV(A_2) & \Rightarrow & \text{indifferent.} & \end{array} \quad \begin{cases} A_2, & 0 \le p < 0.5, \\ \text{indifferent}, & p = 0.5, \\ A_1, & 0.5 < p \le 1. \end{cases}$$

The sensitivity analysis shows that the coffee shop should keep its current menu when the subjective probability of high traffic is below 50%. When that probability exceeds 50%, launching the dessert line has the higher expected value.

Extended payoff table

Alternative	High traffic S_1 $P(S_1) = p$	Low traffic S_2 $P(S_2) = 1 - p$	Expected Value
A_1: Launch dessert line	40	10	$p(40) + (1-p)(10) = 30p + 10$
A_2: Keep current menu	30	20	$p(30) + (1-p)(20) = 10p + 20$

4 2x2 - Manufacturing company

A small manufacturing company is deciding whether to expand its production capacity or maintain its current production level. The manager is uncertain about market demand during the next quarter. Let p represent the subjective probability of high market demand. Therefore, the probability of low market demand is $1-p$. All payoffs are expressed in thousands of dollars. Using the Expected Value criterion, determine which alternative is preferred for different values of the subjective probability p. Determine whether there is a threshold probability within the interval $0 \le p \le 1$ at which the manager is indifferent between the two alternatives.

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1 - p$
Expand production (A_1)	55	30
Maintain current capacity (A_2)	45	20

4.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Expand production, A_2 = Maintain current capacity

States of nature: S_1 = High demand, S_2 = Low demand

Probabilities: $P(S_1) = p, P(S_2) = 1 - p$ with $0 \le p \le 1$.

Step 2: Calculate the expected value of A_1

For the Expand production alternative:

$$EV(A_1) = p(55) + (1-p)(30)$$

$$= 55p + 30 - 30p = 25p + 30.$$

Thus, the expected payoff from expanding production is $25p + 30$ thousand dollars.

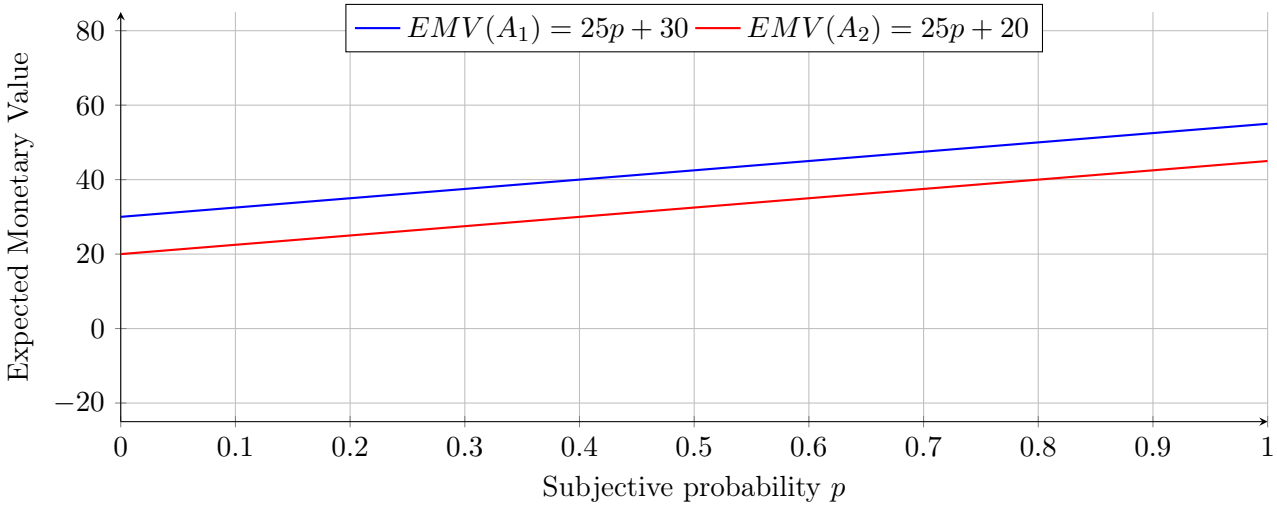
Step 3: Calculate the expected value of A_2

For the Maintain current capacity alternative:

$$EV(A_2) = p(45) + (1-p)(20)$$

$$= 45p + 20 - 20p = 25p + 20.$$

Thus, the expected payoff from maintaining current capacity is $25p + 20$ thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2) \quad \longrightarrow \quad 25p + 30 = 25p + 20$$

$$\longrightarrow \quad 30 = 20.$$

This equation has no solution. Therefore, there is no threshold probability in the interval $0 \leq p \leq 1$ at which the manager is indifferent between the two alternatives.

Step 5: Perform the sensitivity analysis

Compare the expected values:

$$EV(A_1) - EV(A_2) = (25p + 30) - (25p + 20) = 10.$$

Therefore,

$$EV(A_1) > EV(A_2) \qquad \text{for all } 0 \leq p \leq 1.$$

Thus,

$$A_1 \quad \text{is preferred for all} \quad 0 \leq p \leq 1.$$

The sensitivity analysis shows that expanding production has an expected payoff that is always 10 thousand dollars higher than maintaining the current capacity, regardless of the subjective probability of high market demand.

Extended payoff table

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1 - p$	Expected Value
A_1: Expand production	55	30	$p(55) + (1-p)(30) = 25p + 30$
A_2: Maintain current capacity	45	20	$p(45) + (1-p)(20) = 25p + 20$

5 2x2 - Retail company

A retail company is deciding whether to launch a new store in a shopping district or continue operating only through its existing locations. The manager is uncertain about customer demand during the first year. Let p represent the subjective probability of high customer demand. Therefore, the probability of low customer demand is $1 - p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine which alternative is preferred for different values of the subjective probability p. Determine whether there is a threshold probability within the interval $0 \leq p \leq 1$ at which the manager is indifferent between the two alternatives.

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1 - p$
Open new store (A_1)	65	40
Continue existing locations (A_2)	25	35

5.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Open new store, A_2 = Continue existing locations

States of nature: S_1 = High customer demand, S_2 = Low customer demand

Probabilities: $P(S_1) = p, P(S_2) = 1 - p$ with $0 \leq p \leq 1$.

Step 2: Calculate the expected value of A_1

For the Open new store alternative:

$$EV(A_1) = p(65) + (1-p)(40)$$

$$= 65p + 40 - 40p = 25p + 40.$$

Thus, the expected payoff from opening the new store is $25p + 40$ thousand dollars.

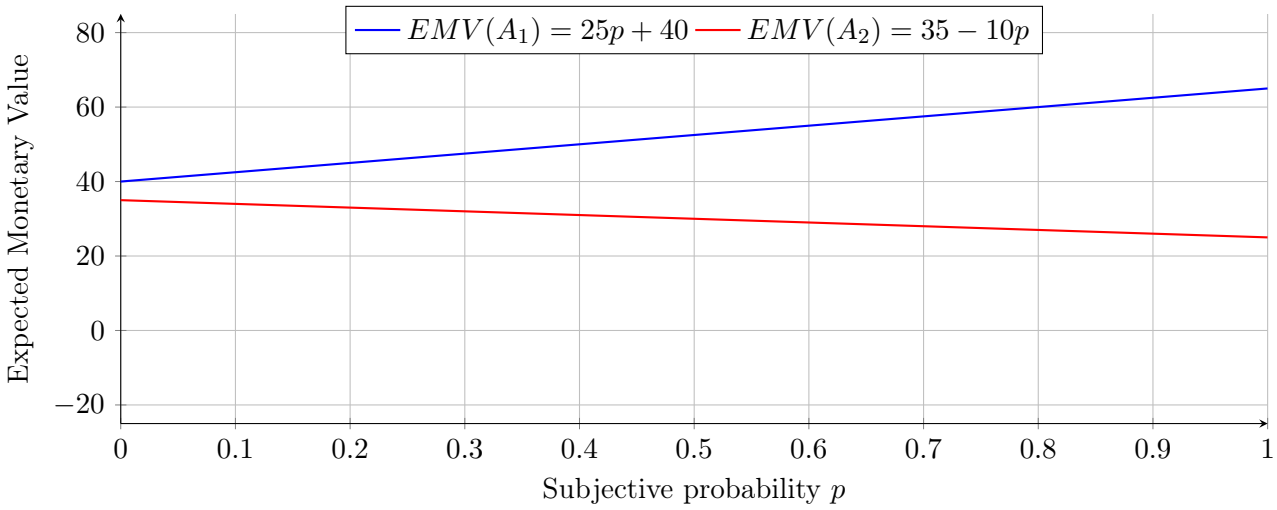
Step 3: Calculate the expected value of A_2

For the Continue existing locations alternative:

$$EV(A_2) = p(25) + (1-p)(35)$$

$$= 25p + 35 - 35p = 35 - 10p.$$

Thus, the expected payoff from continuing with the existing locations is $35 - 10p$ thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2) \quad \longrightarrow \quad 25p + 40 = 35 - 10p$$
$$\longrightarrow \quad 35p = -5 \quad \longrightarrow \quad p = -\frac{1}{7} \approx -0.143.$$

The threshold probability is negative and therefore lies outside the feasible interval $0 \leq p \leq 1$. Thus, there is no threshold probability within the feasible interval at which the manager is indifferent between the two alternatives.

Step 5: Perform the sensitivity analysis

Compare the expected values on the feasible interval.

$$EV(A_1) - EV(A_2) = (25p + 40) - (35 - 10p) = 35p + 5.$$

Since $35p + 5 > 0$ for every $0 \leq p \leq 1$,

$$A_1 \quad \text{is preferred for all} \quad 0 \leq p \leq 1.$$

The sensitivity analysis shows that opening the new store has the higher expected payoff for every feasible subjective probability of high customer demand.

Extended payoff table

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1 - p$	Expected Value
A_1: Open new store	65	40	$p(65) + (1-p)(40) = 25p + 40$
A_2: Continue existing locations	25	35	$p(25) + (1-p)(35) = 35 - 10p$

6 2x2 - Retail company

A retail company is deciding whether to launch a new store in a shopping district or continue operating only through its existing locations. The manager is uncertain about customer demand during the first year. Let p represent the

subjective probability of high customer demand. Therefore, the probability of low customer demand is $1-p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine which alternative is preferred for different values of the subjective probability p. Determine whether there is a threshold probability within the interval $0 \leq p \leq 1$ at which the manager is indifferent between the two alternatives.

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1-p$
Open new store (A_1)	40	65
Continue existing locations (A_2)	35	25

6.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Open new store, A_2 = Continue existing locations

States of nature: S_1 = High customer demand, S_2 = Low customer demand

Probabilities: $P(S_1) = p, P(S_2) = 1-p$ with $0 \leq p \leq 1$.

Step 2: Calculate the expected value of A_1

For the Open new store alternative:

$$EV(A_1) = p(40) + (1-p)(65)$$

$$= 40p + 65 - 65p = 65 - 25p.$$

Thus, the expected payoff from opening the new store is $65 - 25p$ thousand dollars.

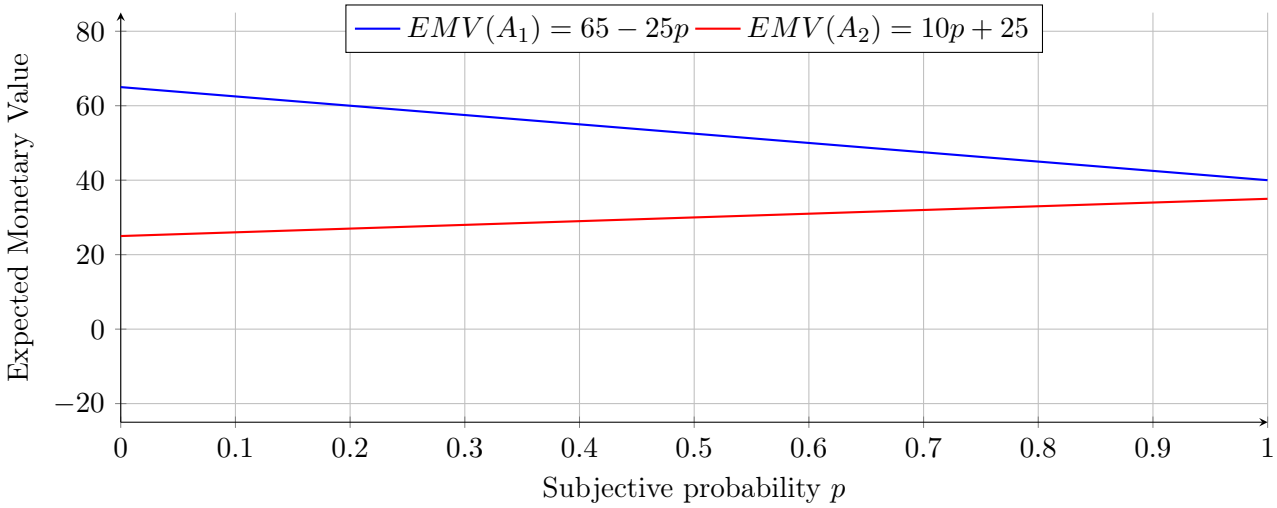
Step 3: Calculate the expected value of A_2

For the Continue existing locations alternative:

$$EV(A_2) = p(35) + (1-p)(25)$$

$$= 35p + 25 - 25p = 10p + 25.$$

Thus, the expected payoff from continuing with the existing locations is $10p + 25$ thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2) \qquad \longrightarrow \qquad 65 - 25p = 10p + 25$$

$$\longrightarrow \qquad 40 = 35p \qquad \longrightarrow \qquad p = \frac{40}{35} = \frac{8}{7} \approx 1.143.$$

The threshold probability is greater than 1 and therefore lies outside the feasible interval $0 \leq p \leq 1$. Thus, there is no threshold probability within the feasible interval at which the manager is indifferent between the two alternatives.

Step 5: Perform the sensitivity analysis

Compare the expected values on the feasible interval.

$$EV(A_1) - EV(A_2) = (65 - 25p) - (10p + 25) = 40 - 35p.$$

Since

$$40 - 35p \geq 5 > 0 \qquad \text{for all } 0 \leq p \leq 1,$$

we have

$$A_1 \quad \text{is preferred for all} \quad 0 \leq p \leq 1.$$

The sensitivity analysis shows that opening the new store has the higher expected payoff for every feasible subjective probability of high customer demand.

Extended payoff table

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1 - p$	Expected Value
A_1: Open new store	40	65	$p(40) + (1-p)(65) = 65 - 25p$
A_2: Continue existing locations	35	25	$p(35) + (1-p)(25) = 10p + 25$

7 2x3 - Delivery company

A delivery company is deciding whether to expand its delivery fleet or continue operating with its current fleet. The manager is uncertain about the level of demand during the next year. Demand may be high, moderate, or low.

Let p represent the subjective probability of high demand. The probability of moderate demand is $2p$, while the probability of low demand is $1 - 3p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine which alternative is preferred for different values of the subjective probability p. Identify the threshold probability at which the manager is indifferent between the two alternatives.

Alternative	High demand S_1 $P(S_1) = p$	Moderate demand S_2 $P(S_2) = 2p$	Low demand S_3 $P(S_3) = 1 - 3p$
Expand fleet (A_1)	80	55	20
Keep current fleet (A_2)	50	45	35

7.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Expand fleet, A_2 = Keep current fleet

States of nature: S_1 = High demand, S_2 = Moderate demand, S_3 = Low demand

Probabilities:

$$P(S_1) = p, \qquad P(S_2) = 2p, \qquad P(S_3) = 1 - 3p.$$

Since all probabilities must be nonnegative,

$$1 - 3p \geq 0 \qquad \Longrightarrow \qquad p \leq \frac{1}{3}.$$

Therefore, the feasible interval is

$$0 \leq p \leq \frac{1}{3}.$$

Step 2: Calculate the expected value of A_1

For the Expand fleet alternative:

$$EV(A_1) = p(80) + 2p(55) + (1 - 3p)(20)$$

$$= 80p + 110p + 20 - 60p = 130p + 20.$$

Thus, the expected payoff from expanding the fleet is $130p + 20$ thousand dollars.

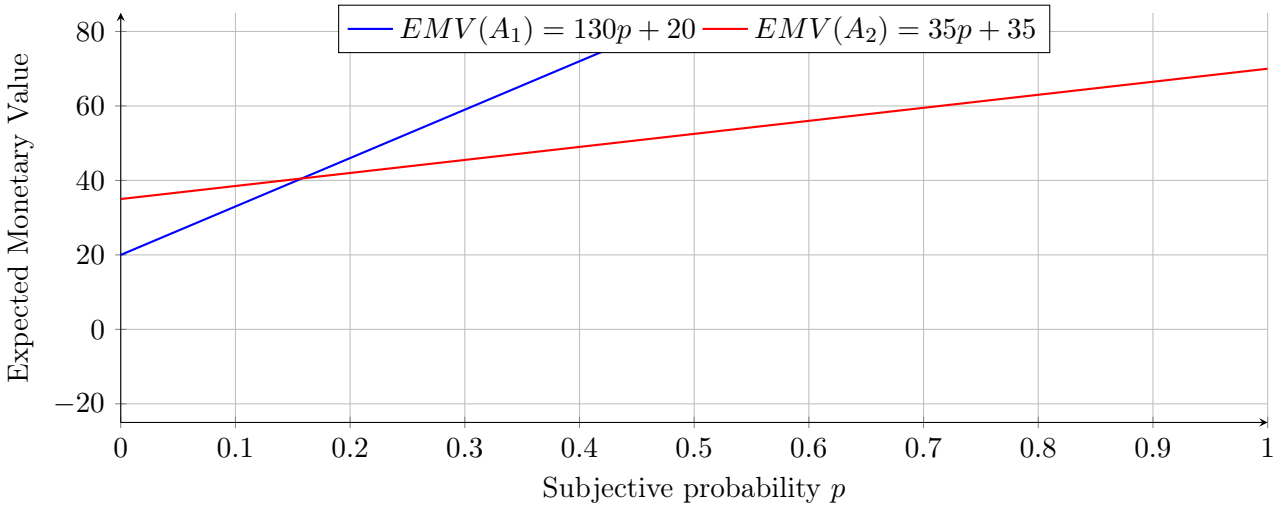
Step 3: Calculate the expected value of A_2

For the Keep current fleet alternative:

$$EV(A_2) = p(50) + 2p(45) + (1 - 3p)(35)$$

$$= 50p + 90p + 35 - 105p = 35p + 35.$$

Thus, the expected payoff from keeping the current fleet is $35p + 35$ thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2) \qquad \longrightarrow \qquad 130p + 20 = 35p + 35$$

$$\longrightarrow \qquad 95p = 15 \qquad \longrightarrow \qquad p = \frac{15}{95} = \frac{3}{19} \approx 0.158.$$

Since

$$0 < \frac{3}{19} < \frac{1}{3},$$

the threshold lies within the feasible interval.

Step 5: Perform the sensitivity analysis

Compare the expected values:

$$EV(A_1) - EV(A_2) = (130p + 20) - (35p + 35) = 95p - 15.$$

Therefore,

$$\begin{aligned} &A_2 \text{ is preferred if } 0 \le p < \frac{3}{19}, \\ &A_1 \text{ and } A_2 \text{ are equivalent if } p = \frac{3}{19}, \\ &A_1 \text{ is preferred if } \frac{3}{19} < p \le \frac{1}{3}. \end{aligned}$$

The sensitivity analysis shows that keeping the current fleet is preferred when the subjective probability of high demand is relatively low. Once p exceeds $\frac{3}{19}$, expanding the fleet has the higher expected payoff.

Extended payoff table

Alternative	High demand S_1 $P(S_1) = p$	Moderate demand S_2 $P(S_2) = 2p$	Low demand S_3 $P(S_3) = 1 - 3p$	Expected Value
A_1: Expand fleet	80	55	20	$p(80) + 2p(55) + (1 - 3p)(20) = 13$
A_2: Keep current fleet	50	45	35	$p(50) + 2p(45) + (1 - 3p)(35) = 35$

8 2x3 - Manufacturing company

A manufacturing company is deciding whether to invest in a new automated production system or continue using its current production system. The manager is uncertain about market conditions during the next year. Market conditions may be strong, moderate, or weak.

Let p represent the combined subjective probability of strong or moderate market conditions. The probability of strong market conditions is $\frac{p}{2}$, the probability of moderate market conditions is $\frac{p}{2}$, and the probability of weak market conditions is $1 - p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine which alternative is preferred for different values of the subjective probability p. Identify the threshold probability at which the manager is indifferent between the two alternatives.

Alternative	Strong market S_1 $P(S_1) = \frac{p}{2}$	Moderate market S_2 $P(S_2) = \frac{p}{2}$	Weak market S_3 $P(S_3) = 1 - p$
Automate production (A_1)	90	70	20
Keep current system (A_2)	55	45	40

8.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Automate production, A_2 = Keep current system

States of nature: S_1 = Strong market conditions, S_2 = Moderate market conditions, S_3 = Weak market conditions

Probabilities:

$$P(S_1) = \frac{p}{2}, \qquad P(S_2) = \frac{p}{2}, \qquad P(S_3) = 1 - p.$$

Since

$$\frac{p}{2} + \frac{p}{2} + (1 - p) = 1,$$

and all probabilities must be nonnegative, the feasible interval is

$$0 \leq p \leq 1.$$

Step 2: Calculate the expected value of A_1

For the Automate production alternative:

$$EV(A_1) = \frac{p}{2}(90) + \frac{p}{2}(70) + (1 - p)(20)$$

$$= 45p + 35p + 20 - 20p = 60p + 20.$$

Thus, the expected payoff from automating production is $60p + 20$ thousand dollars.

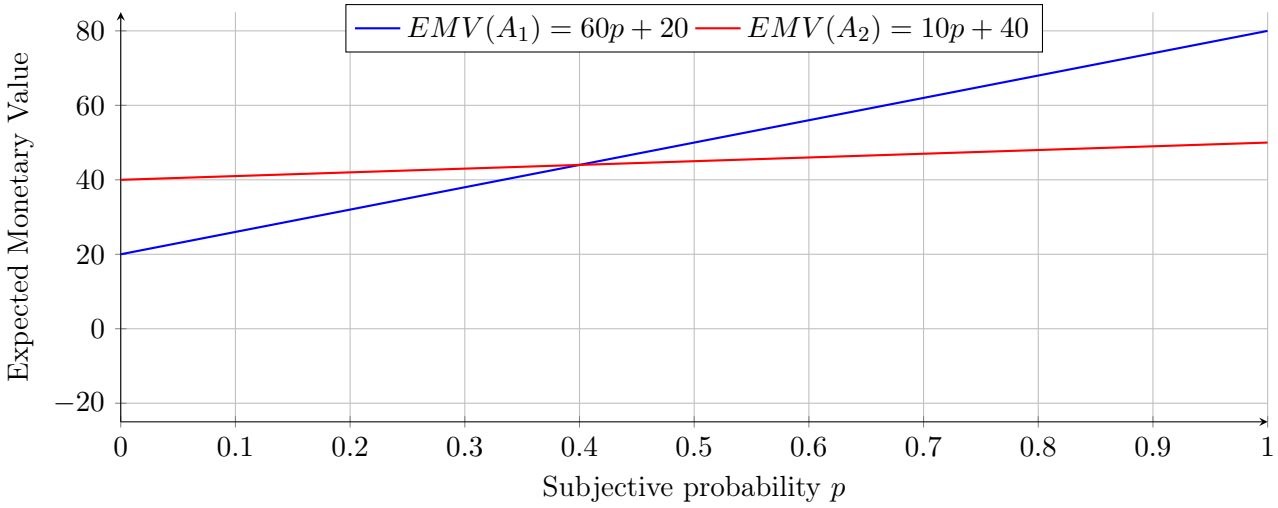
Step 3: Calculate the expected value of A_2

For the Keep current system alternative:

$$EV(A_2) = \frac{p}{2}(55) + \frac{p}{2}(45) + (1 - p)(40)$$

$$= 27.5p + 22.5p + 40 - 40p = 10p + 40.$$

Thus, the expected payoff from keeping the current system is $10p + 40$ thousand dollars.



Step 4: Find the threshold probability

The threshold occurs when the two alternatives have the same expected value:

$$EV(A_1) = EV(A_2) \quad \longrightarrow \quad 60p + 20 = 10p + 40$$

$$\longrightarrow \quad 50p = 20 \quad \longrightarrow \quad p = \frac{2}{5} = 0.4.$$

The threshold lies within the feasible interval $0 \leq p \leq 1$.

Step 5: Perform the sensitivity analysis

Compare the expected values:

$$EV(A_1) - EV(A_2) = (60p + 20) - (10p + 40) = 50p - 20.$$

Therefore,

$$\begin{aligned} &A_2 \text{ is preferred if } 0 \leq p < 0.4, \\ &A_1 \text{ and } A_2 \text{ are equivalent if } p = 0.4, \\ &A_1 \text{ is preferred if } 0.4 < p \leq 1. \end{aligned}$$

The sensitivity analysis shows that keeping the current production system is preferred for lower values of p. Once p exceeds 0.4, investing in the automated production system has the higher expected payoff.

Extended payoff table

Alternative	Strong market S_1 $P(S_1) = \frac{p}{2}$	Moderate market S_2 $P(S_2) = \frac{p}{2}$	Weak market S_3 $P(S_3) = 1 - p$	Expected Value
A_1: Automate production	90	70	20	$\frac{p}{2}(90) + \frac{p}{2}(70) + (1-p)(20) =$
A_2: Keep current system	55	45	40	$\frac{p}{2}(55) + \frac{p}{2}(45) + (1-p)(40) =$

9 3x2 - Technology company

A technology company is deciding which product strategy to adopt for the launch of a new device. The manager is uncertain about the level of demand during the launch period. Let p represent the subjective probability of high demand. Therefore, the probability of low demand is $1 - p$. All payoffs are expressed in thousands of dollars.

Using the Expected Value criterion, determine how the preferred decision changes as the subjective probability p changes. Identify the threshold probabilities at which the manager is indifferent between the alternatives.

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1 - p$
Premium version (A_1)	80	-20
Standard version (A_2)	60	18
Budget version (A_3)	40	25

9.1 Solution

Step 1: Identify the alternatives, states of nature, and probabilities

Alternatives: A_1 = Premium version, A_2 = Standard version, A_3 = Budget version

States of nature: S_1 = High demand, S_2 = Low demand

Probabilities: $P(S_1) = p, P(S_2) = 1 - p$ with $0 \leq p \leq 1$.

Step 2: Calculate the expected value of A_1

For the Premium version alternative:

$$\begin{aligned} EV(A_1) &= p(80) + (1-p)(-20) \\ &= 80p - 20 + 20p \\ &= 100p - 20. \end{aligned}$$

Thus, the expected payoff from the Premium version is $100p - 20$ thousand dollars.

Step 3: Calculate the expected value of A_2

For the Standard version alternative:

$$\begin{aligned} EV(A_2) &= p(60) + (1-p)(18) \\ &= 60p + 18 - 18p \\ &= 42p + 18. \end{aligned}$$

Thus, the expected payoff from the Standard version is $42p + 18$ thousand dollars.

Step 4: Calculate the expected value of A_3

For the Budget version alternative:

$$\begin{aligned} EV(A_3) &= p(40) + (1-p)(25) \\ &= 40p + 25 - 25p \\ &= 15p + 25. \end{aligned}$$

Thus, the expected payoff from the Budget version is $15p + 25$ thousand dollars.

The thresholds occur when two alternatives have the same expected value.

Step 5: Find the threshold probabilities

For A_2 and A_3:

$$\begin{aligned} EV(A_2) &= EV(A_3) \\ 42p + 18 &= 15p + 25 \\ 27p &= 7 \\ p &= \frac{7}{27} \approx 0.259. \end{aligned}$$

Step 6: Find the second threshold probability

For A_1 and A_2:

$$\begin{aligned} EV(A_1) &= EV(A_2) \\ 100p - 20 &= 42p + 18 \\ 58p &= 38 \\ p &= \frac{38}{58} = \frac{19}{29} \approx 0.655. \end{aligned}$$

Thus, the manager is indifferent between A_2 and A_3 when the subjective probability of high demand is approximately 25.9%, and indifferent between A_1 and A_2 when it is approximately 65.5%.

Step 7: Perform the sensitivity analysis

Compare the expected values on each side of the thresholds.

$$\begin{aligned} p < 0.259 &\quad \Rightarrow \quad EV(A_3) > EV(A_2), EV(A_1) \quad \Rightarrow \quad \text{choose } A_3, \\ p = 0.259 &\quad \Rightarrow \quad EV(A_2) = EV(A_3) \quad \Rightarrow \quad \text{indifferent between } A_2 \text{ and } A_3, \\ 0.259 < p < 0.655 &\quad \Rightarrow \quad EV(A_2) > EV(A_1), EV(A_3) \quad \Rightarrow \quad \text{choose } A_2, \\ p = 0.655 &\quad \Rightarrow \quad EV(A_1) = EV(A_2) \quad \Rightarrow \quad \text{indifferent between } A_1 \text{ and } A_2, \\ p > 0.655 &\quad \Rightarrow \quad EV(A_1) > EV(A_2), EV(A_3) \quad \Rightarrow \quad \text{choose } A_1. \end{aligned}$$

Therefore, the preferred decision depends on the subjective probability of high demand:

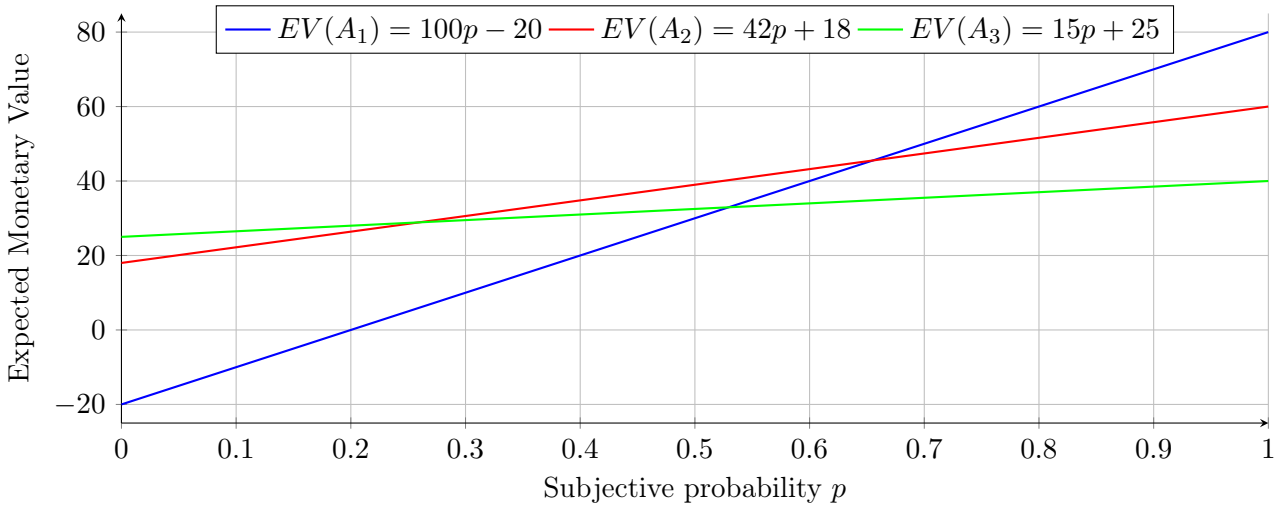
$$\begin{cases} A_3, & 0 \le p < 0.259, \\ \text{indifferent between } A_2 \text{ and } A_3, & p = 0.259, \\ A_2, & 0.259 < p < 0.655, \\ \text{indifferent between } A_1 \text{ and } A_2, & p = 0.655, \\ A_1, & 0.655 < p \le 1. \end{cases}$$

The sensitivity analysis shows that the company should choose the Budget version when the subjective probability of high demand is below approximately 25.9%. Between 25.9% and 65.5%, the Standard version has the highest expected value. When the probability of high demand exceeds 65.5%, the Premium version becomes the preferred decision.

Extended payoff table

Alternative	High demand S_1 $P(S_1) = p$	Low demand S_2 $P(S_2) = 1 - p$	Expected Value
A_1: Premium version	80	-20	$p(80) + (1-p)(-20) = 100p - 20$
A_2: Standard version	60	18	$p(60) + (1-p)(18) = 42p + 18$
A_3: Budget version	40	25	$p(40) + (1-p)(25) = 15p + 25$

Expected Value graph



10 3x2 - Renewable energy company

A renewable energy company must choose one of three solar-farm designs. The return depends on whether electricity demand is high (S_1) or low (S_2). Let $P(S_1) = p$ and $P(S_2) = 1 - p$. Payoffs are in thousands of dollars.

Alternative	High demand S_1	Low demand S_2
Large solar farm (A_1)	90	-30
Medium solar farm (A_2)	60	15
Small solar farm (A_3)	35	30

Using the Expected Value criterion, determine the preferred design for each value of p and identify the threshold probabilities.

10.1 Solution

The expected-value functions are

$$EV(A_1) = 120p - 30, \qquad EV(A_2) = 45p + 15, \qquad EV(A_3) = 5p + 30.$$

The relevant intersections are

$$EV(A_2) = EV(A_3) \Rightarrow p = \frac{3}{8}, \qquad EV(A_1) = EV(A_2) \Rightarrow p = \frac{3}{5}.$$

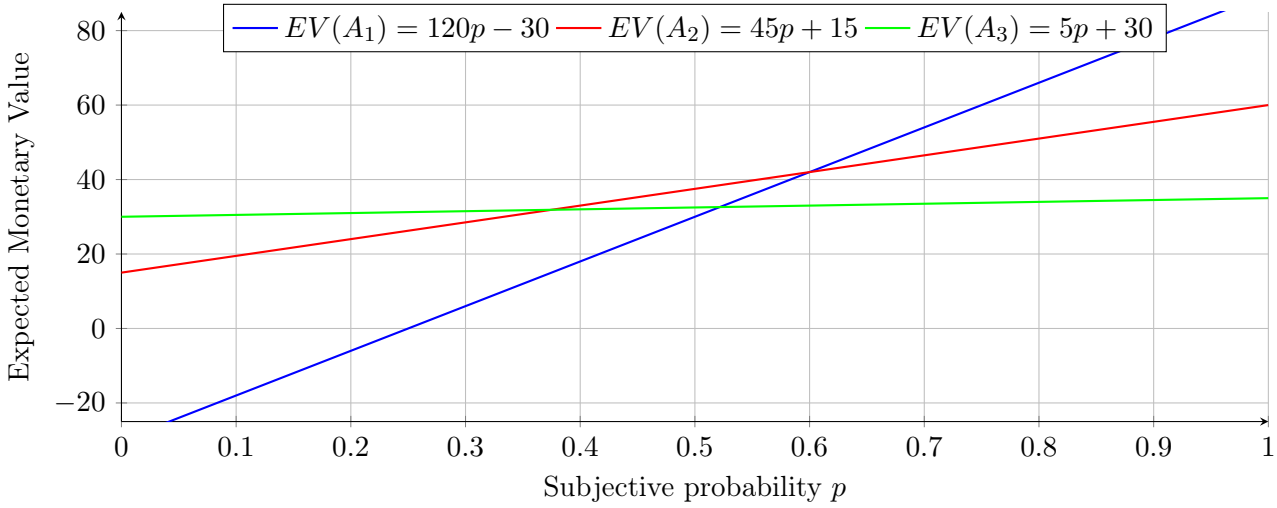
(The intersection of A_1 and A_3 is not a decision threshold because A_2 has a greater expected value there.) Therefore,

$$\begin{cases} A_3, & 0 \le p < \frac{3}{8}, \\ \text{indifferent between } A_2 \text{ and } A_3, & p = \frac{3}{8}, \\ A_2, & \frac{3}{8} < p < \frac{3}{5}, \\ \text{indifferent between } A_1 \text{ and } A_2, & p = \frac{3}{5}, \\ A_1, & \frac{3}{5} < p \le 1. \end{cases}$$

Extended payoff table

Alternative	S_1 (p)	S_2 ($1-p$)	Expected Value
A_1: Large	90	-30	$120p - 30$
A_2: Medium	60	15	$45p + 15$
A_3: Small	35	30	$5p + 30$

Expected Value graph



11 3x2 - Restaurant chain

A restaurant chain is choosing a format for a new location. Customer traffic may be high (S_1) or low (S_2), with probabilities p and $1 - p$, respectively. Payoffs are in thousands of dollars.

Alternative	High traffic S_1	Low traffic S_2
Flagship restaurant (A_1)	110	-40
Neighborhood restaurant (A_2)	70	20
Pop-up restaurant (A_3)	45	35

Using the Expected Value criterion, determine the preferred format for each value of p and identify the threshold probabilities.

11.1 Solution

The expected-value functions are

$$EV(A_1) = 150p - 40, \qquad EV(A_2) = 50p + 20, \qquad EV(A_3) = 10p + 35.$$

The relevant intersections are

$$EV(A_2) = EV(A_3) \Rightarrow p = \frac{3}{8}, \qquad EV(A_1) = EV(A_2) \Rightarrow p = \frac{3}{5}.$$

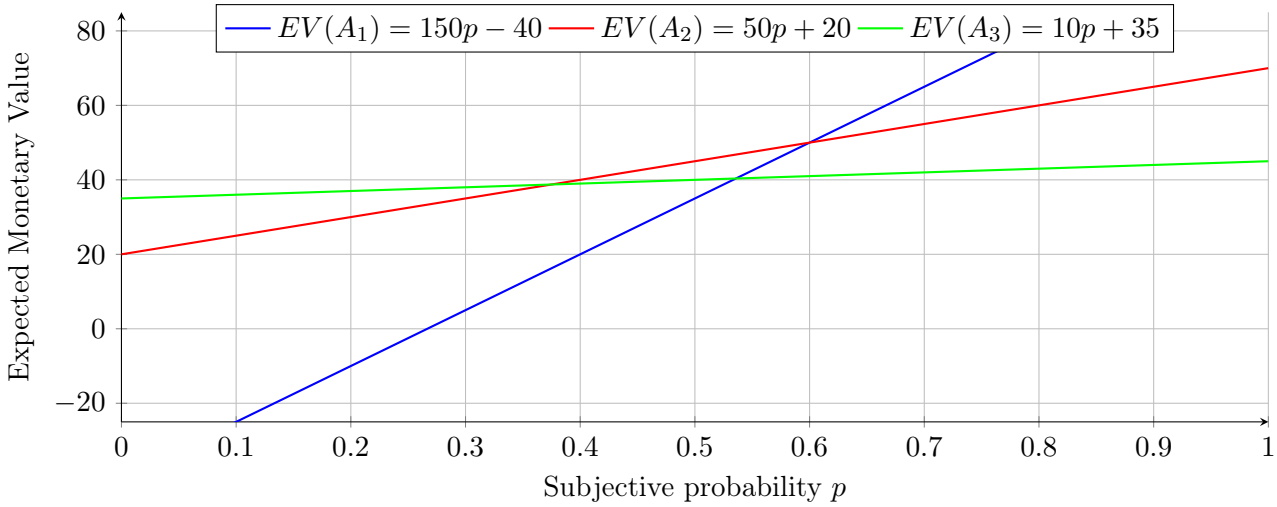
Although A_1 and A_3 also intersect, A_2 has a greater expected value at that probability. The decision rule is

$$\begin{cases} A_3, & 0 \le p < \frac{3}{8}, \\ \text{indifferent between } A_2 \text{ and } A_3, & p = \frac{3}{8}, \\ A_2, & \frac{3}{8} < p < \frac{3}{5}, \\ \text{indifferent between } A_1 \text{ and } A_2, & p = \frac{3}{5}, \\ A_1, & \frac{3}{5} < p \le 1. \end{cases}$$

Extended payoff table

Alternative	S_1 (p)	S_2 $(1-p)$	Expected Value
A_1: Flagship	110	-40	$150p - 40$
A_2: Neighborhood	70	20	$50p + 20$
A_3: Pop-up	45	35	$10p + 35$

Expected Value graph



12 3x2 - Agricultural cooperative

An agricultural cooperative must choose a planting strategy before it knows whether the growing season will be favorable (S_1) or unfavorable (S_2). Let $P(S_1) = p$ and $P(S_2) = 1 - p$. Payoffs are in thousands of dollars.

Alternative	Favorable S_1	Unfavorable S_2
High-intensity plan (A_1)	95	-25
Standard plan (A_2)	65	15
Resilient plan (A_3)	40	30

Using the Expected Value criterion, determine the preferred planting plan for each value of p and identify the threshold probabilities.

12.1 Solution

The expected-value functions are

$$EV(A_1) = 120p - 25, \qquad EV(A_2) = 50p + 15, \qquad EV(A_3) = 10p + 30.$$

The relevant intersections are

$$EV(A_2) = EV(A_3) \Rightarrow p = \frac{3}{8}, \qquad EV(A_1) = EV(A_2) \Rightarrow p = \frac{4}{7}.$$

The intersection of A_1 and A_3 is not a decision threshold because A_2 has a greater expected value there. Thus,

$$\begin{cases} A_3, & 0 \le p < \frac{3}{8}, \\ \text{indifferent between } A_2 \text{ and } A_3, & p = \frac{3}{8}, \\ A_2, & \frac{3}{8} < p < \frac{4}{7}, \\ \text{indifferent between } A_1 \text{ and } A_2, & p = \frac{4}{7}, \\ A_1, & \frac{4}{7} < p \le 1. \end{cases}$$

Extended payoff table

Alternative	S_1 (p)	S_2 $(1-p)$	Expected Value
A_1: High-intensity	95	-25	$120p - 25$
A_2: Standard	65	15	$50p + 15$
A_3: Resilient	40	30	$10p + 30$

Expected Value graph

